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Assignment - 3

Q1. Double Number Pyramid

for n = 4

```
1
1 2
1 2 3
1 2 3 4
1 2 3
1 2
1
```

* Dry Run

n = 4

i = 1 ; i<=n
↳ j = 1 ; j<=i
↳ 1

i = 2
↳ j<=i
↳ 1 2

i = 3
↳ j<=i
↳ 3

i = n-1 ; i>=1
↳ j=1 ; j<=i
↳ 1 2 3

```
1 1
1 2 1
1 2 3
1 2 3 4
```

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Number Butterfly

Q2

```
1       1
1 2     2 1
1 2 3   3 2 1
1 2 3 4 4 3 2 1
1 2 3 4 4 3 2 1
1 2 3   3 2 1
1 2     2 1
1       1
```

* Dry Run n = 4

* 1.02

i = 1 ; i<=n.
↳ j = 1 ; j<=i
↳ 1

2<=n
j<=i
↳ 1 2

* ~~right~~ spacing

for i ; j<=2*(n-i)
↳ .

Right
* j = i ; j>=1
↳ 1

i>=1
1

Q3 Cross (x) Number Pattern

```
1   1
 2 2
  3
 4 4
5   5
```

n = 5

i = 1; i <= n

↳ j = 1; j <= n

↳ if(j == i || j == n-i+1){

↳ " "

j <= n

↳ (else) " 2 " 2

Q4. Spiral Layer Pattern

```
3 3 3 3 3
3 2 2 2 3
3 2 1 2 3
3 2   2 3
3 3 3 3 3
```

• Dry Run n = 3;

i = 1 ; i <= 2*n-1

↳ j = 1; j <= 2*n-1

↳

Q6 Diamond with continous numbers

n = 4

```
   1
  2 3
 4 5 6
7 8 9 10
 11 12 13
  14 15
   16
```

n = 4 , num = 1

Dry run

i = 1 ; i <= n
↳
j = 1 ; j <= i
↳ num
↳ ,
numtt

2 <= i
↳ num
↳ 2 3
numtt

i = n-1 ; i >= 1
↳ j = 1 ; j <= i
↳ num
numtt

Q7. Binary Diamond

```
   1
  0 1
 1 0 1
0 1 0 1
 1 0 1
  0 1
```

```
   1
  0 1
 1 0 1
0 1 0 1
 1 0 1
  1
```

Dry Run n = 4

i = 1 ; i <= n
↳
val = i % 2
↳ j = 1 ; j <= i
↳ val + val = 1 - val
1 + 1 = 1 - 1

2 <= n
↳ i <= i
↳ val = 2 % 2 = 0
↳ 0

1 % 2 = 1